

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
 $KE = 1\text{ eV} = m_e v^2 / 2$
 $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$
 $m_e v^2 = 2 \times 1.6 \times 10^{-19}\text{ J}$
 $v = \sqrt{2 \times 1.6 \times 10^{-19} / m_e}$
 $h = 6.626 \times 10^{-34}\text{ Js}$, $m_e = 9.11 \times 10^{-31}\text{ kg}$
 $\lambda = h / v = h / \sqrt{2 \times 1.6 \times 10^{-19} / m_e}$
 $\lambda = 5.27 \times 10^{-9}\text{ m} = 5.27 \times 10^{-10}\text{ cm}$
</think><answer>
F. 5.27×10^{-9} m
</answer>

